

Name: _____

MARKING KEY

Date: _____



Year 10 Mathematics

Test 1, 2014

Topics – Surds & Indices

Total Time:

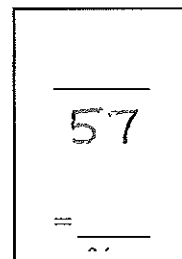
60 minutes

Weighting:

6% of the year.

Equipment
Allowed:

½ page notes (A4 one side)



CALCULATOR FREE TEST

1. [2 marks: 1, 1]

Write these surds in simplest form:

a) $\sqrt{28}$

$$\begin{aligned} & \sqrt{4 \times 7} \\ &= 2\sqrt{7} \end{aligned}$$

b) $\sqrt{300}$

$$\begin{aligned} & \sqrt{100 \times 3} \\ &= 10\sqrt{3} \end{aligned}$$

2. [2 marks: 1, 1]

Simplify each of the following

a) $2\sqrt{7} - 6\sqrt{7} + 9\sqrt{7}$

$$= 5\sqrt{7}$$

b) $2\sqrt{5} + 8\sqrt{14} - 8\sqrt{5} - 6\sqrt{14}$

$$= -6\sqrt{5} + 2\sqrt{14}$$

3. [3 marks: 1, 2]

Simplify each of the following

a) $\sqrt{3} + \sqrt{20}$

$$\begin{aligned} & \sqrt{3} + \sqrt{4 \times 5} \\ &= \sqrt{3} + 2\sqrt{5} \end{aligned}$$

b) $\sqrt{8} + \sqrt{24} + \sqrt{18} + \sqrt{54}$

$$\begin{aligned} & \sqrt{4 \times 2} + \sqrt{4 \times 6} + \sqrt{9 \times 2} + \sqrt{9 \times 6} \\ &= 2\sqrt{2} + 2\sqrt{6} + 3\sqrt{2} + 3\sqrt{6} \\ &= 5\sqrt{2} + 5\sqrt{6} \end{aligned}$$

working ①
answer ①

4. [3 marks: 1, 1, 1]

Simplify each of the following

a) $\sqrt{3} \times \sqrt{3}$

$$= 3$$

b) $\sqrt{5} \times \sqrt{12}$

$$\begin{aligned} &= \sqrt{60} \\ &= \sqrt{4 \times 15} \\ &= 2\sqrt{15} \end{aligned}$$

c) $5\sqrt{7} \times 2\sqrt{8}$

$$\begin{aligned} &= 10\sqrt{56} \\ &= 10 \times \sqrt{4 \times 14} \\ &= 10 \times 2\sqrt{14} \\ &= 20\sqrt{14} \end{aligned}$$

5. [2 marks: 1, 1]
Simplify each of the following

a) $\sqrt{27} \div \sqrt{2}$

$$= \frac{\sqrt{9}\sqrt{3}}{\sqrt{2}}$$

$$= \frac{3\sqrt{3}}{\sqrt{2}}$$

b) $\frac{18\sqrt{75}}{3\sqrt{3}}$

$$= \frac{6\sqrt{25}\sqrt{3}}{\sqrt{3}}$$

$$= 6 \times 5$$

$$= 30$$

6. [4 marks: 1, 3]
Simplify each of the following:

a) $3(5 + \sqrt{3})$

$$= 15 + 3\sqrt{3}$$

b) $(4 + \sqrt{5})(\sqrt{2} - \sqrt{8})$

$$= 4\sqrt{2} - 4\sqrt{8} + \sqrt{10} - \sqrt{40}$$

$$= 4\sqrt{2} - 4\sqrt{4}\sqrt{2} + \sqrt{10} - \sqrt{4}\sqrt{10}$$

$$= 4\sqrt{2} - 4 \times 2\sqrt{2} + \sqrt{10} - 2\sqrt{10}$$

$$= 4\sqrt{2} - 8\sqrt{2} + \sqrt{10} - 2\sqrt{10}$$

$$= -6\sqrt{2} - \sqrt{10}$$

working ②
answer ①

7. [4 marks: 1, 1, 1, 1]
Evaluate

a) $\frac{3^6}{3^3} = 3^{6-3} = 3^3 = 27$

b) $2^{-4} = \frac{1}{2^4} = \frac{1}{16}$

c) $4^0 + 2(x + b^2)^0$

$$= 1 + 2 \times 1$$

$$= 1 + 2$$

$$= 3$$

d) $(2^3)^2$

$$= 2^6$$

$$= 64$$

8. [10 marks: 1, 1, 2, 2, 2, 2]
Simplify each of the following.

a) $3x^3 \times 4x^2y^4 \times 2xy^5$

$$= 24x^{3+2+1}y^{4+5}$$

$$= 24x^7y^9$$

b) $(2x^2y^3)^2$

$$= 2^2x^{2 \times 2}y^{3 \times 2}$$

$$= 4x^4y^6$$

c) $\frac{80x^8y^{16}}{12x^2y^9}$

$$= \frac{20x^{8-2}y^{16-9}}{3}$$

$$= \frac{20x^6y^7}{3}$$

d) $(x^5y^2)^4 \times (2x^2y^6)^3$

$$= x^{5 \times 4}y^{2 \times 4} \times 2^3x^{2 \times 3}y^{6 \times 3}$$

$$= x^{20}y^8 \times 8x^6y^{18}$$

$$= 8x^{20+6}y^{8+18}$$

$$= 8x^{26}y^{26}$$

working (1)
answer (1)

e) $\frac{5x^4 \times 2x \times \frac{1}{2}y^9}{15x^2y^3}$

$$= \frac{5x^{4+1}y^9}{15x^2y^3}$$

$$= \frac{1x^5y^9}{3x^2y^3}$$

$$= \frac{x^{5-2}y^{9-3}}{3}$$

$$= \frac{x^3y^6}{3}$$

f) $\frac{40x^9}{y} \div \frac{12x}{2y^4}$

$$= \frac{40x^9}{y} \times \frac{2y^4}{12x}$$

$$= \frac{40x^9y^4}{6xy}$$

$$= \frac{20x^{9-1}y^{4-1}}{3}$$

$$= \frac{20x^8y^3}{3}$$

9. [4 marks: 2, 2]

Simplify each of the following. Write your answers with positive indices.

a) $7x^7y^{-2} \times 8x^{-5}y^{-4}$

$$= 56x^{7+(-5)}y^{-2+(-4)}$$

$$= 56x^2y^{-6}$$

$$= \frac{56x^2}{y^6}$$

working ①
answer ①

b) $\frac{m^{-2}n^{-5}}{m^{-8}n^3}$

$$= m^{-2-(-8)}n^{-5-3}$$

$$= m^6n^{-8}$$

$$= \frac{m^6}{n^8}$$

10. [4 marks: 2, 2]

Simplify each of the following. Write your answers with positive indices

a) $\left(\frac{3m^{-6}n^4}{p^4q^{-2}}\right)^{-3}$

$$= \frac{3^{-3}m^{-6 \times -3}n^{4 \times -3}}{p^{4 \times -3}q^{-2 \times -3}}$$

$$= \frac{3^{-3}m^{18}n^{-12}}{p^{-12}q^6}$$

$$= \frac{m^{18}p^{12}}{3^3n^{12}q^6}$$

working ①
answer ①

b) $\frac{3a^4b^{-5}}{(2x^{-2}y)^{-3}}$

$$= \frac{3a^4b^{-5}}{2^{-3}x^{-2 \times -3}y^{-3}}$$

$$= \frac{3a^4b^{-5}}{2^{-3}x^6y^{-3}}$$

$$= \frac{3 \times 2^3 a^4 b^{-5}}{x^6 y^{-3}}$$

$$= \frac{3 \times 8 a^4 y^3}{x^6 b^5}$$

$$= \frac{24 a^4 y^3}{x^6 b^5}$$

11. [6 marks: 3, 3]

Simplify each of the following. Write your answers with positive indices.

a) $\left(\frac{2b^2c^4}{5b^{-4}c^{-6}}\right)^{-2}$

$$= \frac{2^{-2}b^{2 \times -2}c^{4 \times -2}}{5^{-2}b^{-4 \times -2}c^{-6 \times -2}}$$

$$= \frac{2^{-2}b^{-4}c^{-8}}{5^{-2}b^8c^{12}}$$

$$= \frac{2^{-2}b^{-4-8}c^{-8-12}}{5^{-2}}$$

$$= \frac{2^{-2}b^{-12}c^{-20}}{5^{-2}}$$

$$= \frac{5^2}{2^2b^{12}c^{20}}$$

$$= \frac{25}{4b^{12}c^{20}}$$

working (2)
answer (1)

b) $\left(\frac{-3m^0n^5}{n^{-2}p^{-2}}\right)^3$

$$= \frac{(-3)^3m^{0 \times 3}n^{5 \times 3}}{n^{-2 \times 3}p^{-2 \times 3}}$$

$$= \frac{-27m^0n^{15}}{n^{-6}p^{-6}}$$

$$= \frac{-27n^{15-(-6)}}{p^{-6}}$$

$$= \frac{-27n^{21}}{p^{-6}}$$

$$= -27n^{21}p^6$$

12. [4 marks: 2, 2]

Simplify each of the following. Write your answers in simplest root form

a) $a^{\frac{2}{3}} \times a^{\frac{4}{5}}$

$$= a^{\frac{2}{3} + \frac{4}{5}}$$

$$= a^{\frac{10}{15} + \frac{12}{15}}$$

$$= a^{\frac{22}{15}}$$

$$= \sqrt[15]{a^{22}}$$

working (1)
answer (1)

b) $\left(a^{\frac{1}{3}}b^{\frac{1}{5}}\right)^3$

$$= a^{\frac{1}{3} \times 3} b^{\frac{1}{5} \times 3}$$

$$= ab^{\frac{3}{5}}$$

$$= a\sqrt[5]{b^3}$$

13. [6 marks: 3, 3]

Simplify each of the following. Write your answers with positive indices.

a) $\left(\frac{4a^6}{b^5}\right)^3 \div \left(\frac{4^2 a^2}{4b^{-2}}\right)^4$

$$= \frac{4^3 a^{6 \times 3}}{b^{5 \times 3}} \div \frac{4^{2 \times 4} a^{2 \times 4}}{4^4 b^{-2 \times 4}}$$

$$= \frac{4^3 a^{18}}{b^{15}} \div \frac{4^8 a^8}{4^4 b^{-8}}$$

$$= \frac{4^3 a^{18}}{b^{15}} \times \frac{4^4 b^{-8}}{4^8 a^8}$$

$$= \frac{4^{3+4} a^{18+(-8)} b^{-8-15}}{4^8}$$

$$= \frac{4^7 a^{10} b^{-23}}{4^8}$$

$$= \frac{a^{10}}{4 b^{23}}$$

working (2)
answer (1)

b) $\frac{\left(x^{\frac{1}{2}} y^{\frac{1}{3}} z^{\frac{1}{4}}\right)^2}{\left(x^{\frac{1}{3}} y^{\frac{2}{3}} z^{\frac{1}{4}}\right)^{\frac{3}{2}}}$

$$= \frac{x^{\frac{1}{2} \times 2} y^{\frac{1}{3} \times 2} z^{\frac{1}{4} \times 2}}{x^{\frac{1}{3} \times \frac{3}{2}} y^{\frac{2}{3} \times \frac{3}{2}} z^{-\frac{1}{4} \times \frac{3}{2}}}$$

$$= \frac{x^1 y^{\frac{2}{3}} z^{\frac{1}{2}}}{x^{\frac{1}{2}} y^1 z^{\frac{3}{8}}}$$

$$= x^{1 - \frac{1}{2}} y^{\frac{2}{3} - 1} z^{\frac{1}{2} - \frac{3}{8}}$$

$$= x^{\frac{1}{2}} y^{\frac{-1}{3}} z^{\frac{1}{8}}$$

14. [3 marks]

Evaluate the following for $x = 4$

$$3x^{-2} \times \left(\frac{2}{x}\right)^4 \div \frac{3x}{(x^2 y^3)}$$

$$= 3(4)^{-2} \times \left(\frac{2}{4}\right)^4 \div \frac{3(4)}{(4^2)^3}$$

$$= 3(4^{-2}) \times \frac{2^4}{4^4} \times \frac{4^6}{3 \times 4}$$

$$= \frac{3 \times 4^{-2+6}}{4^4 \times 3 \times 4} \times 2^4$$

$$= \frac{4^4 \times 2^4}{4^4 \times 4} = \frac{4^4 \times 2^4}{4^5} = 4^{-1} \times 2^4 = \frac{2^4}{4} = \frac{16}{4} = 4$$

working (2)
answer (1)

~ END OF TEST ~

64

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15. [5 marks: 1, 2, 2]

Rationalise the denominator:

$$\begin{aligned} \text{a)} \quad & \frac{1}{\sqrt{11}} \times \frac{\sqrt{11}}{\sqrt{11}} \\ &= \frac{\sqrt{11}}{11} \end{aligned}$$

$$\begin{aligned} \text{b)} \quad & \frac{8\sqrt{7}}{3\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} \\ &= \frac{8\sqrt{14}}{3 \times 2} \\ &= \frac{8\sqrt{14}}{6} \\ &= \frac{4\sqrt{14}}{3} \end{aligned}$$

$$\begin{aligned} \text{c)} \quad & \frac{5\sqrt{5}}{\sqrt{3}-3} \times \frac{\sqrt{3}+3}{\sqrt{3}+3} \\ &= \frac{5\sqrt{15} + 15\sqrt{5}}{3-9} \\ &= \frac{5\sqrt{15} + 15\sqrt{5}}{-6} \end{aligned}$$

15. [2 marks]

Expand and simplify:

$$(\sqrt{4} + 3\sqrt{5})^2$$

$$(\sqrt{4} + 3\sqrt{5})(\sqrt{4} + 3\sqrt{5})$$

$$\begin{aligned} &= 4 + 3\sqrt{20} + 3\sqrt{20} + 9\sqrt{25} \\ &= 4 + 6\sqrt{20} + 9 \times 5 \\ &= 4 + 6\sqrt{4 \times 5} + 45 \\ &= 4 + 6 \times 2\sqrt{5} + 45 \\ &= 49 + 12\sqrt{5} \end{aligned}$$